works in Nanjing. We have been involved in ultra-precision machining of millimeter-wave waveguide structures, co

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TERAHERTZ COMMUNICATION FOR 6G NETWORKS IN NANJING.

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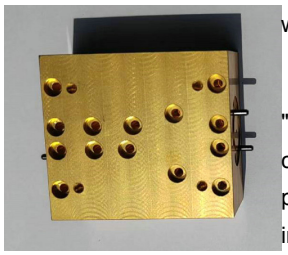
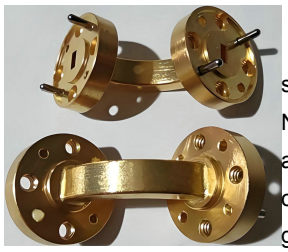
For decades, we have customized millimeter-wave waveguides, terahertz structural components, and sub-millimeter wave cavities for universities in Nanjing. Our rich experience ensures quality. Inquiries are welcome.

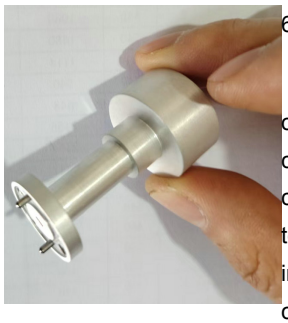


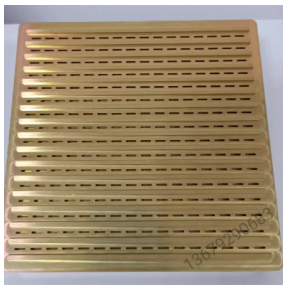
As the first city in China to systematically build a "6G City" through top-level design, Nanjing is forging a complete path from leading core technologies and pioneering application scenarios to a closed-loop industrial system through its innovative practices of forward-looking布局, hardcore scientific research, full-chain transformation, and city-wide collaboration. With the strong momentum of regional innovation, Nanjing supports the national 6G strategy in moving from blueprint to reality. The city is showcasing its latest achievements in 6G technology R&D, scenario implementation, and ecosystem construction to the world. Holding core 6G technologies, Nanjing has achieved numerous world-first results.



While global 6G technology is still in the early stages of vision research and standard exploration, Nanjing has continuously secured one "world first" after another in key 6G technologies through sustained original innovation. It has become an irreplaceable core source of 6G R&D globally, which is the greatest confidence behind Nanjing's ambition to build a "6G City."

For developing the 6G industry, Nanjing possesses unique innovation foundations and platform advantages. The city has gathered a number of top-tier universities and research institutions in the field of information and communications, with the national academies consistently targeting the most cutting-edge and critical technological directions. This has laid the talent and platform foundation for Nanjing to lead the world in 6G technology.





In March 2025, Nanjing established a dedicated office to promote the next-generation information and communications industry, systematically coordinating citywide resources to advance 6G vision research, technological breakthroughs, and application integration, providing organizational support for the construction of the "6G City."

Impressive technological breakthroughs have secured Nanjing's foothold in the global 6G race. To date, the Terahertz Laboratory has achieved over a dozen "world firsts" in the 6G field, produced more than 30 major innovations, and twice won the National Science and Technology Award. In 2022, the laboratory globally released the first 100–200 Gb/s terahertz wireless transmission system, setting a world record for wireless communication transmission at the time.

Terahertz communication, millimeter wave waveguides, THz RF Devices, mmWave products for 6G network

Structure Type	Freq. Range	Loss (dB/m)	Features
Hollow Metal (rect./circ.)	30–3000 GHz	<0.1	Low loss, high power, hard to fab

Gap Waveguide	mm-wave to 70 GHz	0.2–0.5	No contact w/ EBG, easy integration
Dielectric (rod, image line)	mm-wave to sub-THz	0.5–2	Surface wave, planar friendly
Folded Waveguide	D-band (110–170 GHz)	N/A (slow-wave)	High impedance, compact
Hollow-core (metal-coated/polymer)	150–3000 GHz	<5	Flexible, metal inner coat
Silicon-based (SIW, ridge)	100–300 GHz	2–5	CMOS compatible, on-chip
Microstrip / CPW	<150 GHz	3–8	High integration, planar
Lens-integrated / GRIN	0.1–10 THz	N/A	High gain, beam collimation
Metasurface / Metamaterial	mm-wave to THz	varies	Wavefront control, smart surface

Just over a year later, the laboratory achieved another technological leap, successfully realizing 1 Tbps real-time wireless transmission in a 6G photonic terahertz communication experiment — a speed 100 times faster than 5G. In layman's terms, downloading an HD movie would take just one second, completely overturning conventional perceptions of mobile communication speed. Even more disruptive, the laboratory has successfully tackled the 6G cell-free communication architecture, breaking the constraints of traditional cellular networks and enabling multiple base stations to work collaboratively. This greatly improves network coverage, system capacity, and resource utilization efficiency, laying a solid technical foundation for Nanjing's "6G City" — with 6G cell-free coverage targeted for 2027.

In terms of key performance indicators, Nanjing’s 6G technology is pushing toward the extreme. The core goal of the Terahertz Laboratory is to achieve 6G air interface transmission latency as low as one hundred microseconds, reliability as high as 99.99999%, and jitter as low as microseconds. This means that in the smart factories of the future, countless industrial devices will be able to coordinate precisely and wirelessly, much like the human nervous system.

At the Global 6G Technology and Industry Ecosystem Conference, the Purple Mountain Laboratories unveiled the world's first 6G outdoor trial network integrating communication, sensing, and intelligence. In November, they also released the industry's first "endogenous security network resilience evaluation and testing service platform for 6G" and the "endogenous security high-performance interconnect prototype system for 6G AI Edge computing."

To advance 6G, it is essential to learn from the experience of 5G development — we must not prioritize scale at the expense of quality. Instead, progress should be made through exploration, practice, and continuous improvement, with early trial and error to precisely match industry needs. Nanjing has embraced this philosophy, building a full-chain layout of core 6G technologies — from basic research and technical breakthroughs to standard setting and security protection. With tangible innovation outcomes, Nanjing has demonstrated that regional innovation is the cornerstone of 6G technological breakthroughs.

In Nanjing, the "6G+X" scenario matrix serves as a key driver. 6G technology has already entered multiple fields, including the low-altitude economy, industrial manufacturing, urban governance, smart healthcare, and ecological protection. A number of tangible and perceptible real-world application scenarios have been implemented, bringing future communication technology into the city's development and citizens' daily lives. Nanjing has developed typical application scenarios around three core directions: 6G + industrial internet, 6G + smart city, and 6G + low-altitude economy. The city plans to build at least 50 vertical 6G application scenarios by 2027. Currently, four core demonstration scenarios have fully entered the trial-and-first-use stage.

In the field of public services, Nanjing Hospital of Traditional Chinese Medicine has created a "5G-A integrated sensing and communication smart healthcare scenario," applying 6G-related sensing and communication technologies to precision diagnosis, remote monitoring, and emergency dispatch, thereby improving the efficiency and quality of medical services. The industrial sector is a core application area for 6G

technology. In the future, 6G will transform the entire smart factory into a coordinated "organic entity," where devices collaborate with microsecond-level precision, fundamentally changing traditional industrial production models. Nanjing is accelerating the industrial application of new 6G technologies and products, such as millimeter-wave/terahertz cell-free communication and deterministic wireless access, injecting strong momentum into the transformation and upgrading of the manufacturing industry.

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